

AlphaGenome

Early Access Program

AI for better understanding the genome

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TEMPUS

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Unifying DNA sequence model that advances **regulatory variant-effect** prediction and promises to **shed new light on genome** **function**



AlphaGenome



Coding variants

gtatata^tcatgatatgtatacgcata^ggatcatgagctca

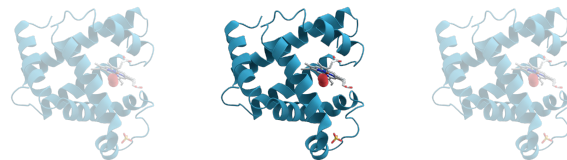


Influence protein function

AlphaMissense

Non-coding variants

gtatata^tcatgatatggatacgcata^ggatcatga^gtca



Influence RNA expression levels

AlphaGenome

Key Capabilities



Long Context & Resolution

Analyzes up to 1 million DNA letters at the resolution of individual letters, crucial for fine-grained details and long-range gene regulation.



Comprehensive Multimodality

Predicts the most diverse range of modalities, providing holistic and in-depth information about the complex steps of gene regulation.



Efficient Variant Scoring

Can efficiently score the impact of a genetic variant on all predicted properties in a single second by contrasting mutated and unmutated sequences.



Splice-Junction Modeling

Explicitly models the location and expression level of RNA splice-junctions directly from sequence, offering deep insights into genetic variant consequences.

Early Access Program



Technical Architecture & API Workflow



Cloud-Native

Vertex AI Model Garden: The model is available via Vertex AI Prediction endpoints, ensuring secure, scalable, and high-throughput processing, governed by Google Cloud Terms.



Standardized Inputs

Client SDK: Accessed via the Python SDK and API, which handles authentication and request formatting seamlessly for EAP users.

Standardized Ontology: Inputs rely on established biological ontologies (UBERON, Cell Ontology) to ensure consistent querying across thousands of potential cell-type and tissue-specific tracks.



Actionable Output

TrackData Objects: Predictions are returned as highly structured `TrackData` objects, containing NumPy arrays of shape `(sequence_length, num_tracks)`.

Rich Metadata: Associated metadata (strand, tissue name) is provided alongside the predictions, enabling easy integration with pandas and specialized visualization tools for analysis.

Early Access Program Timeline

Weeks - 1

Onboarding

Finalize 1-2 research goals, assemble the team, complete all technical setup and validate access to the API.

Weeks 2-3

Exploration

Conducts initial express experiments against the use cases, preferably with verifiable data and validate results. Provide initial feedback.

Weeks 4

Additional experiments

Provide deeper feedback, scale experiments, review the findings to validate the insights provided by AlphaGenome.

Weeks 5+

Post-Engagement

Provide comprehensive feedback to the Google and discuss next steps.

Alpha Genome in Cloud - EAP Details

Scope Requirements

- **Focus:** Target 3-10 research goals
- **Questionnaire:** Complete use-case questionnaire
- **GCP Project:** Set up [Google Cloud](#) project and add users.
- **Service Account:** Create and share a service account, which will be granted access to AlphaGenome API
- **Data:** Ensure appropriate data is available for EAP.
- **Expert:** Secure domain experts for the program who can validate the generated hypothesis and provide high-quality feedback

Terms, Benefits and Expectations

- **Objective :** 6-weeks (no-cost) access to model.
- **Terms:** Governed by our GCP preview [terms](#). HIPAA compliance is not available during EAP. The service is intended for testing and evaluation purposes only. Service & Outputs cannot be commercialized or used in production.
- **API First:** API access is initially provided for integration and testing (with notebook examples). UI to come later.
- **API Throughput:** 2-10 QPS.
- **Support:** Receive direct support from AlphaGenome product team, including **weekly meetings** for feedback and guidance.
- **IP:** Any discoveries, hypotheses, or code generated during the EAP using your proprietary data remain exclusively your IP, fostering a truly collaborative and secure environment for innovation.
- **Commitment:** We expect you'll commit to evaluating the API, provide feedback and be able to share code snippets/examples when needed to reproduce data.

Thank you

